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Synthesis of a new lithium ionic conductor, thio-LISICON–lithium germanium sulfide system

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Abstract

The new thio-LISICON (Lithium SuperIonic CONductor) found in the $\text{Li}_2\text{S}-\text{GeS}_2$, $\text{Li}_2\text{S}-\text{GeS}_2-\text{ZnS}$ and $\text{Li}_2\text{S}-\text{GeS}_2-\text{Ga}_2\text{S}_3$ systems have structures related to the $\gamma\text{-Li}_3\text{PO}_4$ -type. Six new materials were found (Li_2GeS_3 , Li_4GeS_4 , $\text{Li}_2\text{ZnGeS}_4$, $\text{Li}_{4-2x}\text{Zn}_x\text{GeS}_4$, Li_5GaS_4 and $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x\text{S}_4)$, and $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x\text{S}_4)$ showed the highest conductivity of $6.5 \times 10^{-5} \text{ S cm}^{-1}$ at room temperature. Our results on electrochemical measurements indicated that $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x\text{S}_4)$ is stable up to $\approx 5 \text{ V}$ vs. Li. An all-solid-state lithium cells using $\text{Li}/\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x\text{S}_4)/\text{LiCoO}_2$ was also examined. © 2000 Published by Elsevier Science B.V. All rights reserved.

Keywords: Thio-LISICON; Lithium ion conductor; Lithium conductivity; All-solid-state cell

1. Introduction

Lithium ionic conductors are of technological importance for future applications as solid electrolytes for all-solid lithium batteries and may solve the safety problems of the rechargeable lithium ion battery using non-aqueous liquid electrolytes [1]. To improve the stability of the electrolytes, a material search is still necessary, and this includes ceramic crystalline materials. A wide variety of materials have been synthesized in the LISICON [$\text{Li}_{14}\text{Zn}(\text{GeO}_4)_4$] family [2], where the framework is related to the $\gamma\text{-Li}_3\text{PO}_4$ structure, and are formed by GeO_4 , SiO_4 , PO_4 , ZnO_4 , VO_4 tetrahedra or LiO_6 octahedra. A wide range of solid solutions can be

formed by aliovalent substitutions and interstitial lithium ions or lithium vacancies that contribute with their high ionic conductivities at higher temperatures [1]. The conductivity of the orthorhombic solid solution $\text{Li}_{3+x}\text{Si}_x\text{P}_{1-x}\text{O}_4$ ($0 \leq x \leq 0.6$), for example, increased with x or with the number of interstitial lithium ions [3,4]. However, rather low ionic conductivity ($\approx 10^{-6} \text{ S cm}^{-1}$) at room temperature, makes it difficult to use in practical applications. From the conductivity standpoint, the highest value of $10^{-3} \text{ S cm}^{-1}$ at room temperature was obtained for the sulfide glasses in the $\text{Li}_2\text{S}-\text{SiS}_2-\text{LiI}$ [5] and $\text{Li}_2\text{S}-\text{SiS}_2-\text{Li}_3\text{PO}_4$ [6] systems. Sulfides have an advantage in the construction of ionic conductors; larger and more polarizable sulfide ions make the conduction species more mobile.

Crystalline materials might have higher conductivity than the corresponding glasses if their crystal structure is designed for high ionic conduction. For

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example, the highest ionic conductive solids, $\text{Rb}_4\text{Cu}_{16}\text{I}_7\text{Cl}_{13}$ [7,8] and RbAg_4I_5 [9], have much higher ionic conductivity than any copper and silver conducting glasses. Furthermore, another advantage of crystalline materials is their stability at high temperatures. However, inorganic crystalline lithium ion conductors have usually lower conductivities than glasses. A new material search is still necessary to find high lithium ionic conducting crystalline compounds. Only a few materials have been investigated previously in the crystalline sulfides; Li_3PS_4 [10], Li_2SiS_4 and Li_4SiS_4 [11] were reported to have conductivities of 10^{-7} – $10^{-9} \text{ S cm}^{-1}$ at room temperature and no systematic material search has been made.

Here we report a new material family of ceramic lithium ionic conductors.

2. Experimental

The starting materials were Li_2S , GeS_2 , ZnS and Ga_2S_3 (Pure Chemical, >99.9% purity). These were weighed, mixed in appropriate molar ratios in an argon-filled glove box, put into a carbon-coated quartz tube and heated at a reaction temperature of 600–800°C for 8 h. After the reaction, the tube was slowly cooled to room temperature.

X-ray diffraction patterns of the powdered samples were obtained with an X-ray diffractometer (Rigaku RAD-C, 12 kW) with $\text{CuK}\alpha$ radiation. The lattice parameters and structures were refined by Rietveld analysis [12] using the computer program RIETAN-97 β . The diffraction data were collected at each 0.02° step width over a 2θ range from 20 to 110°.

Ionic conductivities were measured by AC impedance methods using a Solatron 1260 frequency response analyzer with a frequency range between 0.1 Hz and 10 MHz. Gold paste was painted onto both side of the samples and treated in an evacuated quartz tube at 350°C for 5 h. The measurements were carried out for the sample with a pellet (8 mm diameter and 2–3 mm thick) between room temperature and 300°C in a nitrogen gas flow. Cyclic voltammetry using the asymmetric cell was examined for the (–) $\text{Li}/\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4/\text{Au}$ cell at room temperature. The electronic conductivity of the sample was measured by the Hebb–Wagner polarization technique [13].

3. Results and discussion

Our material search has been done for a system based on the germanium sulfides (Li_2S – GeS_2 , Li_2S – GeS_2 – ZnS , Li_2S – GeS_2 – Ga_2S_3). Table 1 summa-

Table 1
Materials of the new thio-LISICON family

Compound	Reference	Crystal system	Lattice parameters a, b, c (Å)	Structure type	Conductivity σ (S cm^{-1}) (25°C)
<i>Li_2S–GeS_2 system</i>					
Li_2GeS_3	New compound	Orthorhombic	5.90, 17.95, 6.81	–	9.7×10^{-9} (125°C)
Li_4GeS_4	New compound	Orthorhombic	14.07, 7.75, 6.15	γ - Li_3PO_4	2.0×10^{-7}
<i>Li_2S–GeS_2–ZnS system</i>					
$\text{Li}_2\text{ZnGeS}_4$	New compound	Orthorhombic	7.83, 6.53, 6.21	β - Li_3PO_4	1.4×10^{-9} (50°C)
$\text{Li}_{4-2x}\text{Zn}_x\text{GeS}_4$	New compound	Orthorhombic	14.05, 7.75, 6.16	γ - Li_3PO_4	3.0×10^{-7}
<i>Li_2S–Ga_2S_3 system</i>					
LiGaS_2	JCPDS 42–1267	Orthorhombic	6.51, 7.87, 6.22	β - NaFeO_2	–
Li_5GaS_4	New compound	–	–	–	5.1×10^{-8} (100°C)
<i>Li_2S–GeS_2–Ga_2S_3 system</i>					
$\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4$	New compound	Orthorhombic	6.89, 6.20, 7.96 (at $x = 0.25$)	–	6.5×10^{-5}

rizes materials of the new sulfide LISICON family synthesized in this study. There are seven compounds; six of these are new materials found in this study.

In the $\text{Li}_2\text{S}-\text{GeS}_2$ binary system, new materials were found to exist at the compositions, Li_4GeS_4 and Li_2GeS_3 . Li_4GeS_4 has a diffraction pattern similar to the corresponding oxide LISICON, and its structure was refined using the structural model of $\text{Li}_3\text{Zn}_{0.5}\text{GeO}_4$ [14]. The refinement results are summarized in Table 2. Sulfur atoms are in a hexagonal close packing and cations are distributed over various tetrahedral sites (Fig. 1). Further studies using neutron diffraction measurements are necessary to determine the occupation parameters of these lithium sites. Based on the stoichiometric Li_4GeS_4 , solid solution formations were examined by aliovalent substitutions, $\text{Zn}^{2+} \rightleftharpoons 2\text{Li}^+$ or $\text{Ge}^{4+} \rightleftharpoons \text{Li}^+ + \text{Ga}^{3+}$, which might give lithium vacancies and interstitial lithium ions, respectively, according to the solid

solution formulae: $\text{Li}_{4-x}\text{GeZn}_x\text{S}_4$ and $\text{Li}_{4+x}\text{Ge}_{1-x}\text{Ga}_x\text{S}_4$.

In the ternary $\text{Li}_2\text{S}-\text{GeS}_2-\text{ZnS}$ system, the new compound $\text{Li}_2\text{ZnGeS}_4$ and the solid solution, $\text{Li}_{4-x}\text{GeZn}_x\text{S}_4$ ($0.0 \leq x \leq 0.20$), with the $\gamma\text{-Li}_3\text{PO}_4$ structure were obtained. In the ternary $\text{Li}_2\text{S}-\text{GeS}_2-\text{Ga}_2\text{S}_3$ system, the new solid solution, $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4$, was synthesized. Its monophasic region is indicated in the formation diagram shown in Fig. 2. The region slightly deviates from the tieline between Li_4GeS_4 and Li_5GaS_4 . Fig. 3 shows the X-ray diffraction pattern of $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4$ with $x=0.25$, $\delta=0.025$ and $\delta'=0.14$. The formula indicated is based on a nominal composition used for the sample synthesis. The pattern shows an orthorhombic cell which was found by the auto-indexing program ITO. A wide range of solid solutions were determined to have $0.2 \leq x \leq 0.6$ in $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4$ by the lattice parameter changes.

The conductivity was calculated from the impedance plots. In $\text{Li}_{4.275}\text{Ge}_{0.61}\text{Ga}_{0.25}\text{S}_4$, for example, a semicircle was observed at 25°C for the high-frequency range with a spike in the low-frequency range which is caused by an electrode contribution (Fig. 4). The semicircle was interpreted as the parallel combination of a resistance and a capacitance. The capacitance calculated from the semicircle was of the order of 10^{-12} F which is a typical value for the bulk capacitance. At higher temperatures, no semicircle was observed and only a spike caused by the electrode contribution was observed. The behavior of the impedance plots show characteristics of a pure ionic conductor composed of bulk and electrode contributions where the bulk resistivity values were calculated from the intercept of the spike.

The temperature dependence of the conductivity in $\text{Li}_{4-x}\text{GeZn}_x\text{S}_4$ is shown in Fig. 5(a). The conductivity increases from $x=0.0$ to 0.05 and then decreases with increasing x . The highest conductivity and the lowest activation energy were observed at $x=0.05$. Fig. 5(b) shows the temperature dependence of the conductivity for the $\text{Li}_2\text{S}-\text{GeS}_2-\text{Ga}_2\text{S}_3$ system. The conductivity increases with decreasing x and the highest conductivity was observed for the sample with $x=0.25$ in the $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4$ system. The conductivity value at 25°C was $6.5 \times 10^{-5} \text{ S cm}^{-1}$. As the Arrhenius conductivity curves

Table 2
X-ray Rietveld refinement results for the orthorhombic Li_4GeS_4 ^a

Atom	Site	<i>g</i>	<i>x</i>	<i>y</i>	<i>z</i>
Li(1)	8(d)	1	0.165	0.504	0.675
Li(2)	4(c)	1	0.424	3/4	0.829
Li(3)	8(d)	0.5	0.195	0.203	0.982
Ge	4(c)	1	0.4103(3)	1/4	0.6495(6)
S(1)	8(d)	1	0.3443(3)	0.0135(9)	0.7764(10)
S(2)	4(c)	1	0.0857(5)	3/4	0.7933(12)
S(3)	4(c)	1	0.0613(5)	1/4	0.7292(12)

^a Space group: Pnma, $a=14.0658(2) \text{ \AA}$, $b=7.75102(15) \text{ \AA}$, $c=6.14973(10) \text{ \AA}$, $R_{\text{wp}}=15.24$, $R_p=11.76$, $S=2.31$, $R_1=9.06$, $R_F=4.36$.

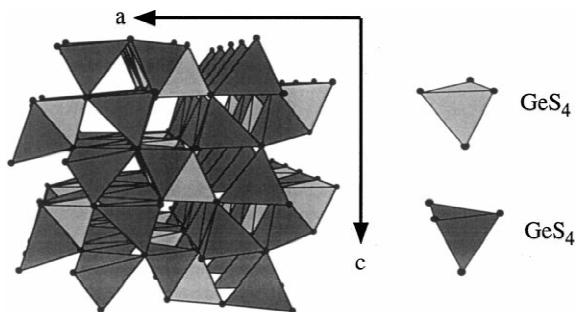


Fig. 1. Structure of Li_4GeS_4 determined by X-ray Rietveld analysis. Although the positions of lithium should be clarified by neutron diffraction data, the framework structure was confirmed. This is a parent structure of the new thio-LISICON family.

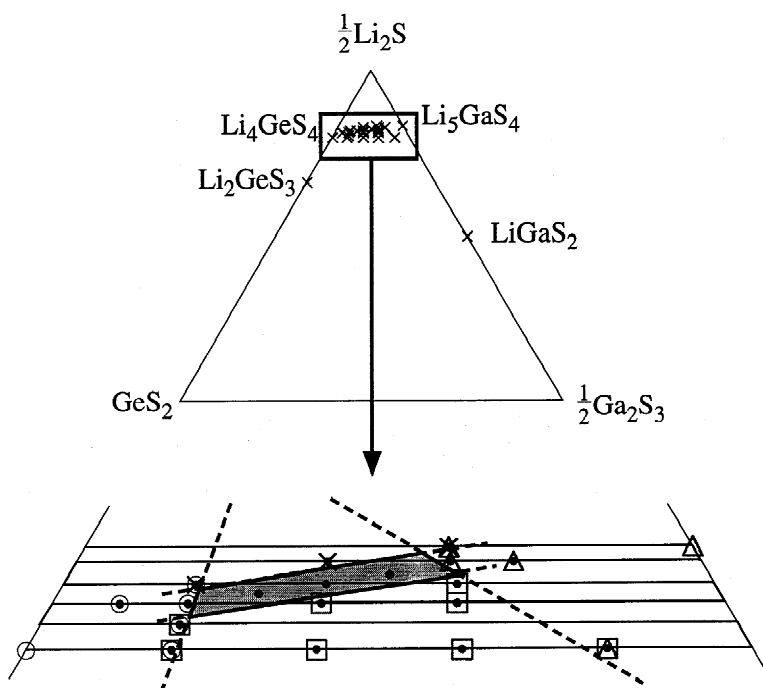


Fig. 2. Formation diagram of the ternary $\text{Li}_2\text{S}-\text{GeS}_2-\text{Ga}_2\text{S}_3$ system. Li_4GeS_4 , Li_2GeS_3 , Li_5GaS_4 and $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4$ are new compounds synthesized in this study.

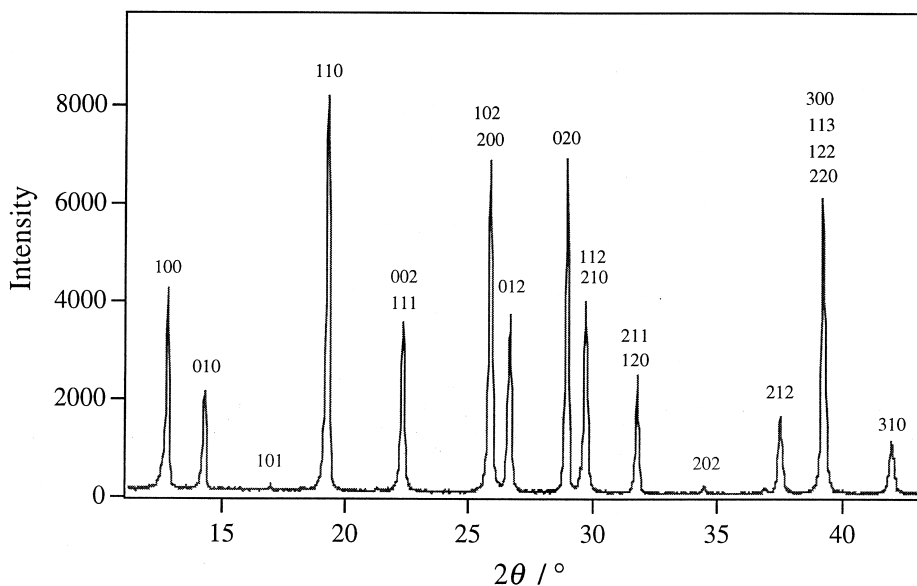


Fig. 3. X-ray diffraction patterns for the orthorhombic $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4$ system. The diffraction peaks are indexed by an orthorhombic cell, $a=6.88$, $b=6.17$, $c=7.94$ Å.

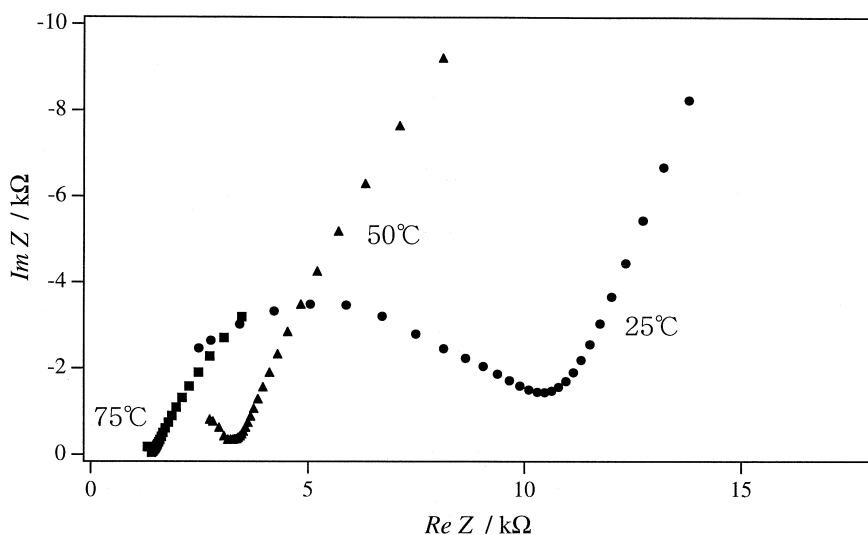


Fig. 4. Impedance plots for $\text{Li}_{4.34}(\text{Ge}_{0.75}\text{Ga}_{0.25})_{0.975}\text{S}_4$.

show gradual change in slope at higher temperatures, the activation energies were calculated for both low temperature and high temperature. The activation energy decreases with increasing x at both high temperatures and low temperatures.

The gradual slope changes are characteristic of typical high-ionic conductors, such as $\text{Rb}_4\text{Cu}_{16}\text{I}_7\text{Cl}_{13}$ [8], the lithium halide spinel (Li_2MCl_4) [15], and PbF_2 [16], and these correspond to a gradual disorder formation in the mobile ion sublattice. The thio-LISICON materials provide new examples of transition from a normal to high ionic conduction state.

The conductivities of the new thio-LISICON materials are higher than those of oxide LISICON ($\text{Li}_{14}\text{Zn}(\text{GeO}_4)_4$) [2] and $\text{Li}_3\text{PO}_4\text{--Li}_4\text{SiO}_4$ solid solutions [3,4] by two or three orders of magnitude, and their values of $6.5 \times 10^{-5} \text{ S cm}^{-1}$ at room temperature are comparable to those of the previous ceramic lithium superionic conductors, H-doped Li_3N ($10^{-3} \text{ S cm}^{-1}$) [17] and $\text{La}_{0.62}\text{Li}_{0.16}\text{TiO}_3$ ($10^{-4} \text{ S cm}^{-1}$) [18].

The chemical stability of the thio-LISICON was examined by the contact of lithium metal with the cold-pressed $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4$. No changes in the X-ray diffraction pattern were observed after 2 weeks for the surface of the pellets, which indicates no surface reaction. To get a first idea of the cycling behavior of the electrolyte under polarizing con-

ditions up to 5 V, cyclic voltammetry using the asymmetric cell was examined in the (–) $\text{Li}/\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4/\text{Au}$ cell at room temperature. The current decreased gradually with cycling between 2 and 5 V, which indicates that no significant decomposition reaction proceeded with increasing cycling. Fig. 6 shows cyclic voltammograms for the (–) $\text{Li}/\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4/\text{Au}$ cell. The scan rate was 10 mV s^{-1} , the electrolyte thickness $L=0.2 \text{ cm}$, and the geometrical area $S=0.28 \text{ cm}^2$. No significant anodic current due to electrolyte decomposition was observed up to 5 V. This indicates that the electrolyte has chemical and electrochemical stability. The electronic conductivity of the sample was measured using the Hebb–Wagner polarization technique [13]. Fig. 7 shows Hebb–Wagner polarization curves obtained from the (–) $\text{Li}/\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4/\text{Au}$ cell at room temperature. Each voltage point within Fig. 7 represents a complete chronoamperometric experiment, which was terminated when a stationary state was reached. The total electronic conductivity ($\sigma_e + \sigma_h$) changes at around 1.7 V, and it is assumed that for voltages greater than 1.7 V, the total electronic conductivity adjacent to the irreversible side is mainly attributed to holes. The conductivity of electrons and holes can be separately determined from the slope and the axial intersection using the following equation:

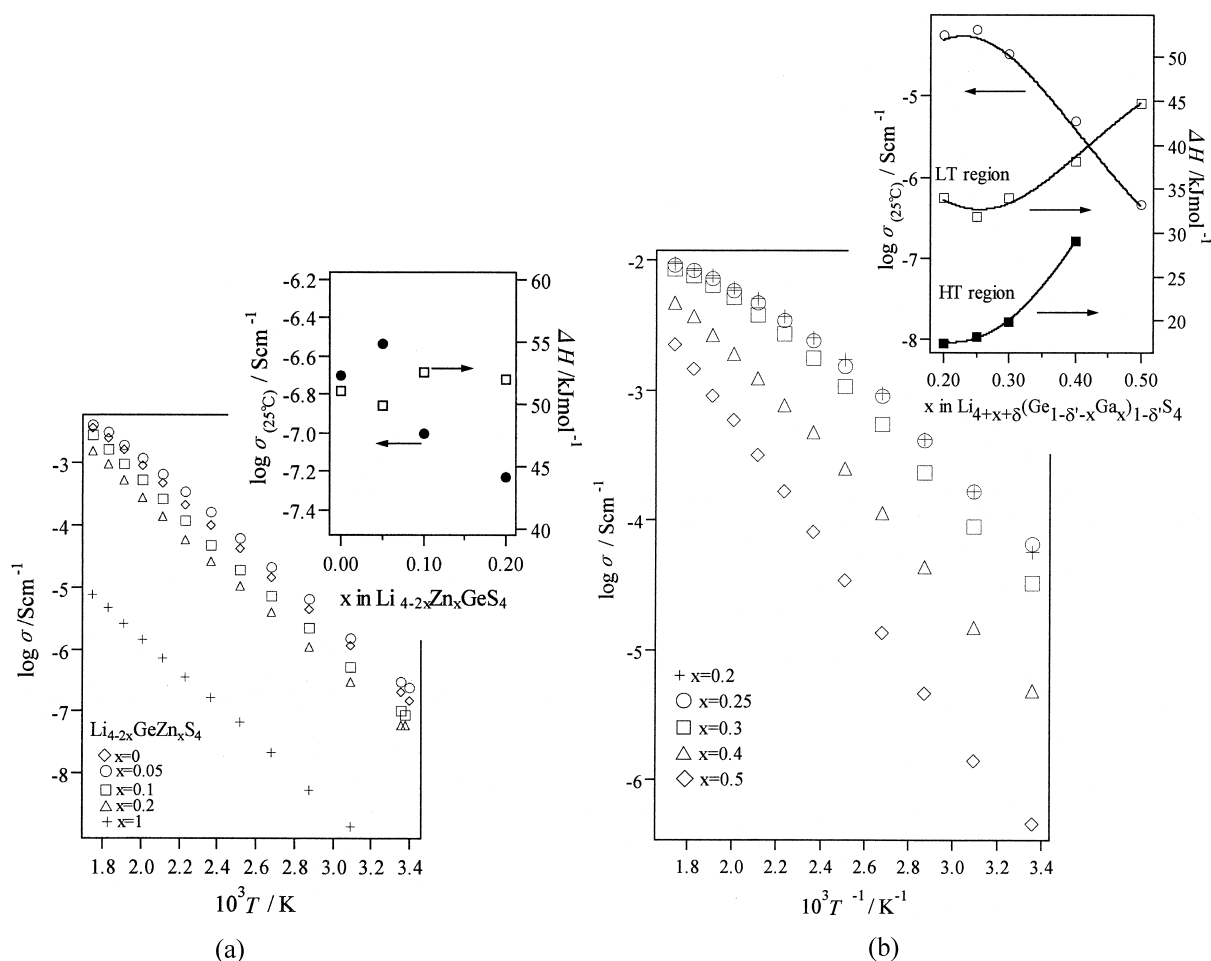


Fig. 5. Temperature dependence of the conductivity for (a) $\text{Li}_{4-2x}\text{Zn}_x\text{GeS}_4$ (a), and (b) $\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)_{1-\delta'}\text{S}_4$. Composition dependence of the conductivity and the activation energy is also indicated in the figure.

$$|I_{e,h}|/(e^{EF/RT} - 1) = (RTS/FL)(\sigma_e e^{-EF/RT} + \sigma_h)$$

The value of the hole conductivity σ_h obtained from the linear relations above 1.7 V was $1.7 \times 10^{-10} \text{ S cm}^{-1}$, and the electronic conductivity at an applied voltage of 1.0 V was $6.8 \times 10^{-8} \text{ S cm}^{-1}$. In the present study, σ_e was not obtained. The total conductivity at room temperature was $6.49 \times 10^{-5} \text{ S cm}^{-1}$, and the electronic contribution of the total conductivity was negligibly small.

In/LiCoO₂ electrochemical cells were constructed to investigate the cell performances using the thio-LISICON electrolyte. The electrolyte powder was pressed into a pellet with a diameter of 10 mm and

thickness of 1 mm. The positive electrode material is a mixture of LiCoO₂ and electrolyte powder in a weight ratio of 1:1. It was pressed together with the electrolyte layer into a two-layered pellet. Fig. 8 shows the results for the charge and discharge experiments. The cell was charged and discharged from 0.5 V to 4.2 V at a constant current in the 13–64 $\mu\text{A cm}^{-2}$ range at room temperature. The charge and discharge capacity was about 80 mA h g^{-1} .

With the discovery of thio-LISICON, a wide variety of new materials formed by PS₄, ZnS₄, GeS₄, SiS₄, VS₄, AlS₄ tetrahedra and LiS₆ octahedra are expected, besides the materials synthesized in this

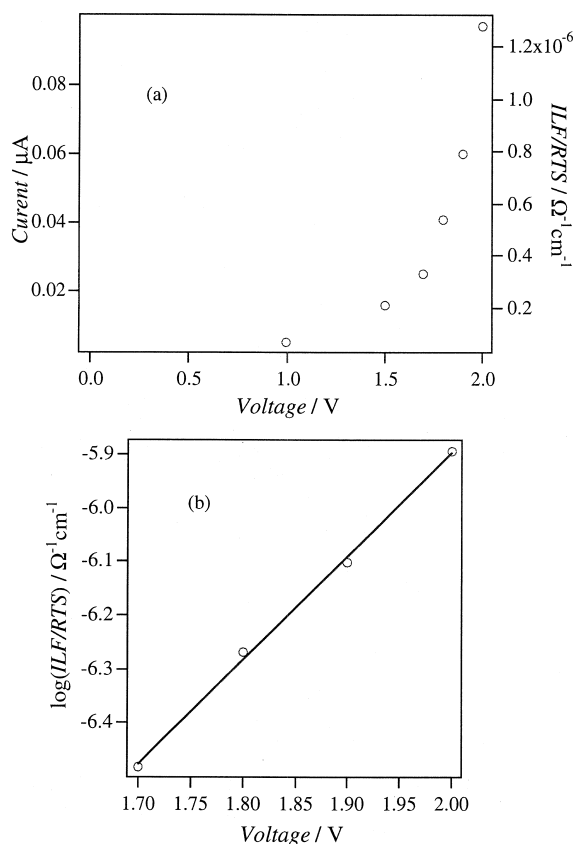


Fig. 6. Cyclic voltammogram of the $\text{Li}/\text{Li}_{4+x+\delta}(\text{Ge}_{1-\delta'-x}\text{Ga}_x)\text{S}_4/\text{Au}$ cell, using a reference Li electrode. The scan rate was 1 mV s^{-1} . No significant current was observed up to 5 V vs. Li. This indicates that the electrolyte was stable for an applied voltage up to 5 V.

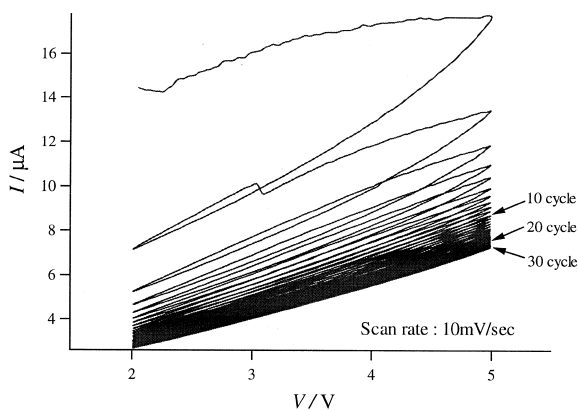


Fig. 7. Wagner polarization curve obtained for the $(-)\text{Li}/\text{Li}_{4.34}(\text{Ge}_{0.75}\text{Ga}_{0.25})_{0.975}\text{S}_4/\text{Au}(+)\text{cell}$ (a). Relationship between $\log(ILF/RTS)$ and the applied voltage (b).

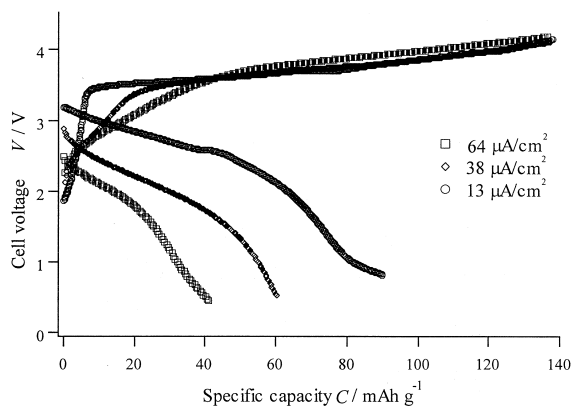


Fig. 8. Charge–discharge curves of the In/LiCoO_2 cell. The current densities were 13, 38 and $64 \mu\text{A cm}^{-2}$.

study. Further studies to optimize these materials in order to obtain a practical solid electrolyte are necessary.

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